

Competitive Exam Practice 7: Microprogrammed Control Unit Design

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CS 2002 Q2.7, 2 marks — verified, see provenance note above*] **[MCQ, 2 marks]**

Horizontal microprogramming:

- (A) does not require use of signal decoders
- (B) results in larger sized microinstructions than vertical microprogramming
- (C) uses one bit for each control signal
- (D) all of the above

Q2 [*GATE CS 2004 Q67, verified — GATEOverflow V3 p.78*] **[MCQ, 2 marks]** The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next-address field (X), and a MUX-select field (Y). There are 8 status bits in the input of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

- (A) 10, 3, 1024
- (B) 8, 5, 256
- (C) 5, 8, 2048
- (D) 10, 3, 512

Q3 [*GATE CS 1996 Q2.25, verified — GATEOverflow V3 p.77*] **[MCQ, 1 mark]** A microprogrammed control unit is required to generate a total of 25 control signals. Assume that during any microinstruction, at most two control signals are active. The minimum number of bits required in the control word to generate the required control signals is:

- (A) 2
- (B) 25
- (C) 10
- (D) 12

- Q4** [*GATE CS 1997 Q5.3, verified — GATEOverflow V3 p.77*] [**MCQ, 1 mark**] A micro-instruction is to be designed to specify: (a) none or one of the three micro-operations of one kind, and (b) none or up to six micro-operations of another kind. The minimum number of bits in the micro-instruction is:
- (A) 9
 - (B) 5
 - (C) 8
 - (D) None of the above
- Q5** [*Original, GATE-pattern*] [**MCQ, 1 mark**] Compared to horizontal microprogramming, vertical microprogramming:
- (A) requires no decoder between control memory and the datapath
 - (B) produces wider control words
 - (C) requires a decoder, adding to the delay before control signals become valid
 - (D) cannot encode mutually exclusive control signals
- Q6** [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a microprogrammed control unit, the component that loads the microprogram counter (MPC) with the starting microaddress of the instruction currently in IR is the:
- (A) Control memory
 - (B) Mapping unit
 - (C) Sequence counter
 - (D) Arithmetic logic unit
- Q7** [*Original, GATE-pattern*] [**MCQ, 2 marks**] Arranged from fastest to slowest per control step, the typical ordering of the following control-unit implementations is:
- (A) Vertical microprogrammed > Horizontal microprogrammed > Hardwired
 - (B) Hardwired > Horizontal microprogrammed > Vertical microprogrammed
 - (C) Horizontal microprogrammed > Hardwired > Vertical microprogrammed
 - (D) Hardwired > Vertical microprogrammed > Horizontal microprogrammed